

Complaint & Feedback Management System

MOBILE APPLICATION · PROJECT REPORT

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DEPARTMENT	Computer Science and Engineering (CSE)
COURSE	Mobile Computing and Application (CSEg3306)
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1 DEEP STUDY & ANALYSIS

1.1 Problem Statement

Many institutions and organizations still manage complaints manually through verbal reports, paper forms, Telegram groups, phone calls, and spreadsheets. This fragmented approach leads to a range of operational and communication failures:

- Complaints get lost in informal communication channels
- Users receive delayed or no responses
- No transparency in complaint tracking or status
- No centralized system or single source of truth
- Poor accountability across departments
- Difficult to generate meaningful reports and analytics

The proposed **Complaint & Feedback Management System** digitizes the entire complaint handling workflow, improving communication, accountability, and transparency between users and administrators.

1.2 User Interviews

Structured interviews were conducted with **10 users** and **2 administrators** across various campus services to identify pain points in the existing complaint process.

User Participants	Admin Participants
• University student	• Student Service Officer
• Dormitory student	• Campus Maintenance Administrator
• Cafeteria customer	
• Staff member	
• Library user	
• ICT service user	
• Campus transportation user	
• Maintenance service user	
• Registrar office visitor	
• Department office visitor	

1.3 Interview Findings

User Pain Points

- Complaints are frequently ignored with no follow-up
- Slow or no responses from responsible offices
- No mechanism to track complaint status
- Repeated submission of the same complaint
- Lack of transparency and clear communication

Administrator Challenges

- Complaints arrive from multiple, uncoordinated channels
- Difficult to organize and prioritize complaint records
- Manual tracking is inefficient and error-prone
- Hard to identify and escalate urgent complaints
- No reporting or analytics tools available

1.4 Real-World Complaint Scenarios

The following scenarios illustrate how the proposed system addresses real problems observed on campus.

Scenario 1

Dormitory Water Shortage

Problem: A student reports a water shortage. Currently routed through a dormitory representative with no tracking.

Solution:

- ✓ Digital complaint submission with real-time status
- ✓ Admin notification and assignment
- ✓ Resolution updates sent to student

Scenario 2

Classroom Projector Failure

Problem: A lecturer reports broken equipment. Currently maintenance requests are forgotten or deprioritized.

Solution:

- ✓ Categorized complaint with image attachment
- ✓ Priority-based management
- ✓ Admin dashboard tracking

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Cafeteria Hygiene Complaint

Problem: Students raise hygiene concerns. Currently spread through social media with no accountability.

Solution:

- ✓ Centralized reporting and complaint history
- ✓ Admin monitoring and response
- ✓ User feedback on resolution

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Internet Connectivity Issue

Problem: Students report poor Wi-Fi. Multiple duplicate complaints, no update mechanism.

Solution:

- ✓ Categorized issue tracking and grouping
- ✓ Technician assignment workflow
- ✓ Status notifications to all affected users

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Registrar Office Delay

Problem: Document processing delays with unclear responsibility and repeated office visits.

Solution:

- ✓ Transparent workflow and responsibility assignment
- ✓ Response tracking system
- ✓ Automated notification system

1.5 As-Is vs. Proposed Workflow

Current (As-Is)	Proposed (To-Be)
User → Representative/Office	User → Mobile Application
→ Administrator (manual)	→ Cloud Database
→ Manual Processing	→ Admin Dashboard
→ Resolution (if reached)	→ Status Update + Notification → Resolution

2 REQUIREMENT SPECIFICATION

2.1 Functional Requirements

Category	User Features	Admin Features
Authentication	Register · Login/Logout · Password Reset	Secure Admin Login
Complaint Mgmt	Submit · Upload · Edit · View History · Search	View All · Categorize · Assign Priority · Respond · Resolve
Tracking	Track Status · Notifications · View Responses	Update Status · Mark Resolved
User Mgmt	Edit Profile	Manage Users · Block · View Activity
Analytics	View Resolution Feedback	Statistics · Monthly Reports · Category Analysis

2.2 Non-Functional Requirements

Attribute	Requirement
Performance	Fast response times, efficient database queries, and scalable architecture to handle growing load.
Security	Secure authentication (JWT), end-to-end data encryption, and role-based access control (RBAC).
Usability	User-friendly interface with responsive design, clear navigation, and minimal learning curve.
Reliability	High system availability, automated backup and recovery, and robust error handling.
Compatibility	Full Android support across multiple screen sizes and device configurations.
Maintainability	Modular, clean-architecture codebase with comprehensive documentation.

2.3 User Stories & Use Cases

Student

- As a student, I want to submit complaints so administrators can resolve my problems.
- As a student, I want to track complaint status so I know the progress.
- As a student, I want notifications so I receive updates immediately.

Administrator

- As an administrator, I want to categorize complaints so they are organized.
- As an administrator, I want to update statuses so users remain informed.

- As an administrator, I want analytics so I can monitor service quality.

3 DESIGN DOCUMENTS

3.1 System Architecture

Layer	Technology	Responsibility
Frontend (Mobile)	Flutter	User-facing mobile application for Android
Backend (API)	FastAPI (Python)	RESTful API, business logic, authentication
Admin Panel	Flutter Web / React	Admin dashboard for complaint management
Database	Cloud Firestore	NoSQL document store for all application data
File Storage	Firebase Storage	Attachments, images, and documents
Notifications	Firebase Cloud Messaging	Push notifications to mobile clients

3.2 Database Schema

The database is organized into five primary entities. Relationships are enforced via ID references.

■ Users

Field	Type
user_id	String (Primary Key)
name	String
email	String
phone	String
role	String (user / admin / staff)
created_at	Timestamp

■ Complaints

Field	Type
complaint_id	String (Primary Key)
user_id	String (FK → Users)
category_id	String (FK → Categories)
title	String
description	String

Field	Type
attachment_url	String
status	String (pending / in_progress / resolved)
priority	String (low / medium / high)
created_at	Timestamp
updated_at	Timestamp

■ Categories

Field	Type
category_id	String (Primary Key)
category_name	String

■ Responses

Field	Type
response_id	String (Primary Key)
complaint_id	String (FK → Complaints)
admin_id	String (FK → Users)
response_message	String
created_at	Timestamp

■ Notifications

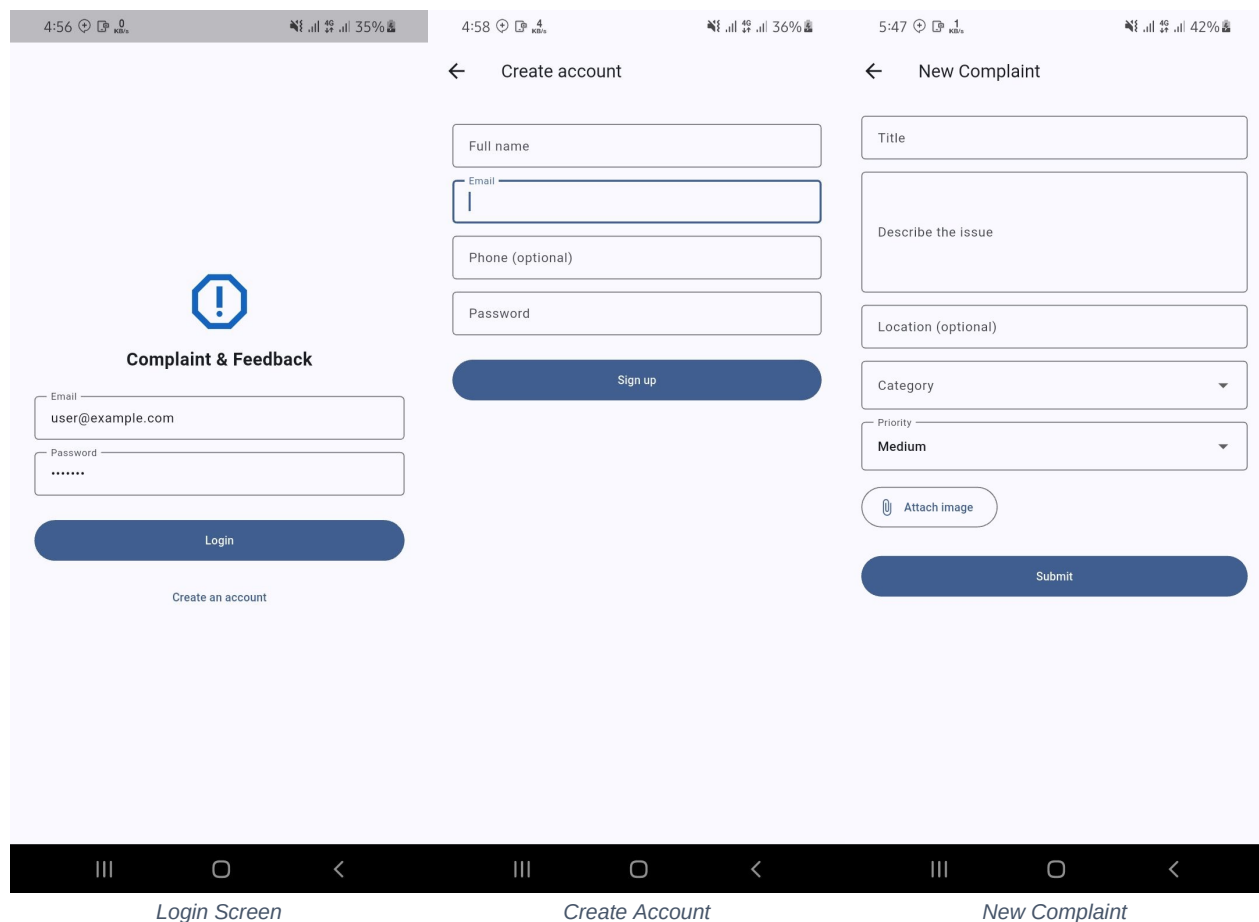
Field	Type
notification_id	String (Primary Key)
user_id	String (FK → Users)
message	String
is_read	Boolean
created_at	Timestamp

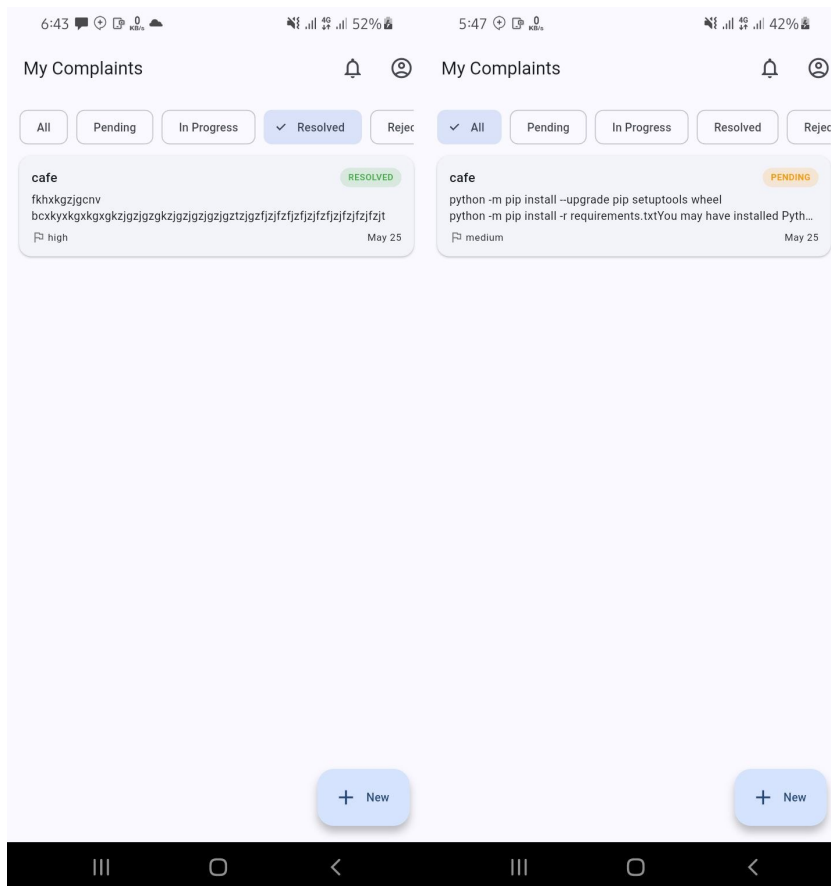
3.3 UI/UX Design

Screen Inventory

User Screens	Admin Screens
Splash Screen	Admin Dashboard
Login Screen	Complaint List
Register Screen	Complaint Detail
Home Dashboard	Category Management
Submit Complaint Screen	Analytics Dashboard
Complaint Detail Screen	User Management
Complaint History Screen	
Notifications Screen	
Profile Screen	

Mobile App Screenshots

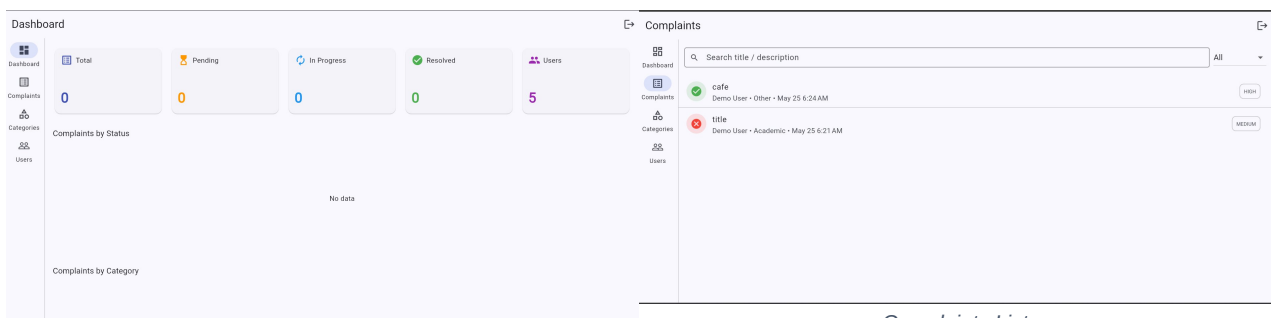




My Complaints

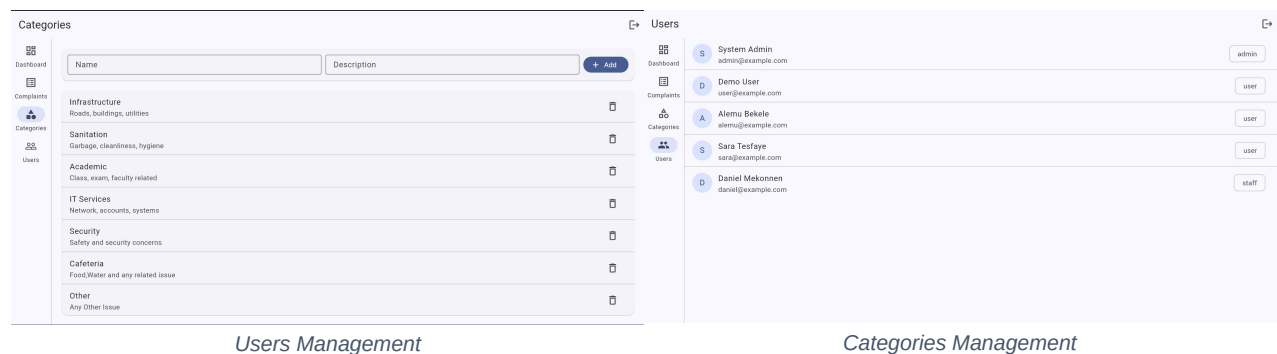
Resolved View

Admin Panel Screenshots



Admin Dashboard

Complaints List



Design Guidelines

- **Primary Color:** Deep Blue (#2563EB) or Purple – used for CTAs, headers, and key actions.
- **Secondary:** White (#FFFFFF) and Light Gray (#F1F5F9) – backgrounds and card surfaces.
- **Status Colors:** Green (resolved) · Amber (pending) · Red (urgent/rejected).
- **Typography:** Poppins (headings) · Inter or Roboto (body text).
- **UX Principles:** Simple navigation, clear status indicators, minimal form fields, real-time updates, accessible design.

3.4 API Design

Authentication

Method	Endpoint	Description
POST	/register	Register a new user account
POST	/login	Authenticate and receive JWT token
POST	/logout	Invalidate the current session token

Complaints

Method	Endpoint	Description
POST	/complaints	Submit a new complaint
GET	/complaints	Retrieve list of complaints (paginated)
GET	/complaints/{id}	Get details of a specific complaint
PUT	/complaints/{id}	Update complaint status or content
DELETE	/complaints/{id}	Remove a complaint

Notifications

Method	Endpoint	Description
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GET	/notifications	Retrieve user notifications
PUT	/notifications/read	Mark notifications as read

4 DEVELOPMENT

4.1 Technology Stack

Technology	Role	Purpose
Flutter	Frontend (Mobile + Web)	Cross-platform UI framework for building natively compiled apps.
FastAPI	Backend (API Server)	High-performance Python web framework for building RESTful APIs.
Cloud Firestore	Database	Scalable NoSQL document database with real-time sync.
Firebase Storage	File Storage	Cloud object storage for user-uploaded attachments and images.
Firebase Cloud Messaging	Push Notifications	Reliable cross-platform notification delivery service.
Git + GitHub	Version Control	Distributed version control with collaborative development workflow.

4.2 Flutter Project Structure

lib/	Root source directory
main.dart	Application entry point
models/	Data models and entity classes
screens/	UI screen widgets
services/	API calls and Firebase services
providers/	State management (Provider)
widgets/	Reusable UI components
utils/	Helper functions and utilities
routes/	Navigation and route definitions
constants/	App-wide constants and themes

4.3 Core Flutter Packages

Package	Description
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firebase_core	Base Firebase SDK initialization
firebase_auth	User authentication with Firebase
cloud_firestore	Firestore database integration
firebase_storage	File upload and download
firebase_messaging	Push notification support
provider	Lightweight state management
image_picker	Camera and gallery image selection
flutter_local_notifications	In-app local notification display

Project Status: This document represents the complete analysis, requirements, design, and development plan for the Complaint & Feedback Management System. The system is designed to significantly improve complaint handling efficiency, transparency, and accountability across the university campus.